Zheng (Thomas) Tang

Senior Deep Learning Engineer at NVIDIA · UW Ph.D. · AI City Challenge Organizing Committee Lead · IEEE T-CSVT Senior Area Editor

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Summary

Senior Deep Learning Engineer at NVIDIA · Former Applied Scientist at Amazon ('19–'21) · UW Ph.D. in Electrical & Computer Engineering · Holder of 19 U.S. patents & author of 28 publications · Developed multimodal fusion search agent for NVIDIA VSS Blueprint with live demo at GTC'26 · Developed Cosmos3 SFT/LoRA workflows and SigLIP2 active learning; drove VLM counting-harness research · Led organization for AI City Challenges at CVPR/ICCV/ECCV engaging 2.5K teams · Built benchmarks on 3D perception & video anomaly reasoning, accepted to NeurIPS'26

Education

University of Washington (UW) — *Seattle, WA, USA*

Ph.D. & M.S. in Electrical & Computer Engineering — *Sep. 2014 - Jun. 2019*

- Ph.D. advisor: Prof. Jenq-Neng Hwang (IEEE Fellow); dissertation: "Robust Video Object Tracking via Camera Self-Calibration"

Beijing University of Posts and Telecommunications & Queen Mary University of London — *Beijing, China*

B.S. in Telecommunications Engineering with Management (Joint Programme) — *Sep. 2010 - Jun. 2014*

- First Class Honours

Work Experience

NVIDIA — *Redmond, WA*

Senior Deep Learning Engineer — *May 2021 - PRESENT*

- Developed **multimodal fusion search agent** for **NVIDIA VSS Blueprint** & post-trained CLIP-based embedding models, with live demo at GTC'26
- Developed **Cosmos3 VLM reranker** post-training with language/vision LoRA, improving retrieval mAP 78% →80% (83% with calibrated fusion)
- Built **DEFT active-learning pipeline** for object attribute search, beating full-data SFT at 81% mAP / 83% Rank-1 with 37% data & 64% less time
- Proposed **reinforcement-learning framework** for object attribute search to optimize hard-query retrieval & ranking upon SFT/DEFT baselines
- Drove **VSS counting-harness research**: Cosmos3-Nano accuracy rose 61% to 69% & Super 67% to 72% on 1K questions with test-fitted routing
- Led **Cosmos physics/geometry benchmark**: 96-question video QA pilot, reference qualification, controlled model evaluation & error analysis
- Prepared **VANTAGE-Bench** & **TAR** benchmarks for physical AI VLM/LLM evaluation, highlighted in the **Cosmos3** release at Computex'26 keynote
- Released **PhysicalAI-SmartSpaces** benchmark (900K downloads) created with **Omniverse**, **Isaac Sim** & **Cosmos** with VLM/LLM auto-labeling
- Created a multi-sensor 3D perception system based on **Sparse4D** for the **Mega Omniverse Blueprint**, with demos at CES'25 & GTC'25 keynotes
- Led decade-long **AI City Challenge** at CVPR/ICCV/ECCV, attracting 2.5K teams across 48 countries, with NVIDIA AI & ECCV Daily News coverage

Amazon — *Seattle, WA*

Applied Scientist — *Jul. 2019 - May 2021*

- Built model-side methods for **Amazon One**, including privacy-aware palm/vein verification & adaptive template updating, and filed 2 patents
- Worked with hardware, product, and service teams to turn biometric ML models into reliable and secure retail and enterprise identity systems

NVIDIA — *Santa Clara, CA*

Intelligent Video Analytics Intern — *Jun. 2018 - Mar. 2019*

- Created **CityFlow**, a city-scale benchmark for multi-target multi-camera (MTMC) vehicle tracking and ReID, accepted to **CVPR'19 (Oral)**
- Proposed **PAMTRI**, a pose-aware multi-task network for vehicle ReID using highly randomized synthetic data, accepted to **ICCV'19**

University of Washington — *Seattle, WA*

Research Assistant — *Jun. 2015 - Jun. 2018*

- Built clustering-based vehicle tracking and camera self-calibration, winning **Track 1** of the **2nd AI City Challenge Workshop** at **CVPR'18**
- Developed multi-camera tracking using visual and semantic features, winning **Track 3** of the **2nd AI City Challenge Workshop** at **CVPR'18**
- Developed a 3D human pose estimator enabling AR overlay in medical simulation under 6-DoF camera motion (funded by ArchieMD Inc.)
- Proposed evolutionary camera self-calibration, a finalist for **Best Scientific & Student Paper Awards** at **ICPR'16** (funded by Prism Skylabs)
- Invented adaptive segmentation & tracking achieving top accuracy (**74.82%**) on **NLPR_MCT** benchmark (funded by Madrona Venture Group)

Professional Services

IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT) — *Remote*

Senior Area Editor — *Jan. 2025 - PRESENT*

AI City Challenge Workshops (CVPR, ICCV, ECCV) — *Remote*

Organizing Committee Lead — *Jun. 2020 - PRESENT*

Selected Publications

The 10th AI City Challenge
Zheng Tang, Shuo Wang, David C. Anastasiu, Ming-Ching Chang, Anuj Sharma, Quan Kong, Munkhjargal Gochoo, Jun-Wei Hsieh, Tomasz Kornuta, Zhedong Zheng, Renran Tian, Judah Goldfeder, Fulgencio Navarro, Yuxing Wang, Yizhou Wang, Sameer Satish Pusegaonkar, Anqi Li, Nalin Dadhich, Ridham Kachhadiya, Dhanishtha Patil, Haoquan Liang, Jiajun Li, Han Zhang, Yilin Zhao, Zaid Pervaiz Bhat, Shuyu Yang, Ashutosh Kumar, Rong Wang, Rafael Martin Nieto, Peter Christiansen, Ahmed Abduljawad, Mohanrasu Shanmugam, Nadeem Shaik, Sujit Biswas, Xunlei Wu, Vidya Murali, Rama Chellappa
Proc. ECCV Workshops, 2026, Malmö, Sweden

PSI-VQA: Evaluating multimodal LLM social reasoning under ambiguous pedestrian intent in automated driving
Shaozhi Wang, Yin-Chun Lu, Zheng Tang, Han Zhang, David C. Anastasiu, Renran Tian
Proc. ECCV Workshops, 2026, Malmö, Sweden

Physical AI Smart Spaces: A large-scale benchmark for multi-camera 3D perception in smart spaces
Yuxing Wang, Yizhou Wang, Anqi Li, Shuo Wang, Sameer Satish Pusegaonkar, Haoquan Liang, Jiajun Li, Shenxin Jiang, Jianhe Yuan, Shangru Li, Tongwei Dai, Zihao Chen, David C. Anastasiu, Sujit Biswas, Xunlei Wu, Zheng Tang
Proc. NeurIPS, 2026, Atlanta, GA, USA

From detection to understanding – a multi-task dataset for traffic anomaly reasoning
Han Zhang, Yilin Zhao, Zaid Pervaiz Bhat, Zheng Tang, Varun Praveen, Vidya Nariyambut Murali, David C. Anastasiu, Tomasz Kornuta
Proc. NeurIPS, 2026, Atlanta, GA, USA

The staged knowledge distillation in video classification: Harmonizing student progress by a complementary weakly supervised framework
Chao Wang, Zheng Tang
T-CSVT 34.8 (2024) pp. 6646–6660. 2024

WSSGCN: Wide sub-stage graph convolutional networks
Chao Wang, Zheng Tang, Hailu Xu
Neurocomputing 602 (2024) p. 128273. 2024

Radiance field learners as UAV first-person viewers
Liqi Yan, Qifan Wang, Junhan Zhao, Qiang Guan, Zheng Tang, Jianhui Zhang, Dongfang Liu
Proc. ECCV, pp. 88–107, 2024, Milan, Italy

PAMTRI: Pose-aware multi-task learning for vehicle re-identification using highly randomized synthetic data
Zheng Tang, Milind Naphade, Stan Birchfield, Jonathan Tremblay, William Hodge, Ratnesh Kumar, Shuo Wang, Xiaodong Yang
Proc. ICCV, pp. 211–220, 2019, Seoul, Korea

CityFlow: A city-scale benchmark for multi-target multi-camera vehicle tracking and re-identification
Zheng Tang, Milind Naphade, Ming-Yu Liu, Xiaodong Yang, Stan Birchfield, Shuo Wang, Ratnesh Kumar, David C. Anastasiu, Jenq-Neng Hwang
Proc. CVPR, pp. 8797–8806, 2019, Long Beach, CA, USA

Selected Patents

Segment-based video reasoning
Cheng-Yen Yang, Anqi Li, Yuxing Wang, Zheng Tang, Sujit Biswas
U.S. Patent Application No. 19/563,882, 2026

Context-aware contrastive learning
Cheng-Yen Yang, Zheng Tang, Sameer Satish Pusegaonkar, Yuxing Wang, Varun Praveen, Sujit Biswas
U.S. Patent Application No. 64/119,912, 2026

Updating identification data in automated user-identification systems
Zheng Tang, Lior Zamir, Prithviraj Banerjee, Manoj Aggarwal, Gerard Medioni, Dilip Kumar
U.S. Patent No. 11,816,932 B1, 2023

Honors & Awards

2025 **Invited Keynote Speaker,** UrbanAI Workshop at NeurIPS'25 — *San Diego, CA, USA*
2024 **Best Presentation Award & Best Poster Award,** NTECH'24 (NVIDIA's annual internal technical conference) — *Santa Clara, CA, USA*
2022 **T-CSVT Best Associate Editor Award,** IEEE Circuits and Systems Society (CASS) — *Remote*
2019 **People's Choice Award,** Code for the Kingdom (C4TK) Hackathon — *Seattle, WA, USA*
2016 **Finalist for Best Scientific Paper Award & Best Student Paper Award,** ICPR'16 — *Cancún, México*

Skills

Tech Stack	Python, C/C++, Java, JavaScript, MATLAB; PyTorch, NumPy, PEFT, vLLM, TensorRT, Omniverse, Isaac Sim
Model Training	SFT, reinforcement learning, LoRA, active learning, targeted data creation, VLM/LLM-assisted data curation
Multimodal AI	CLIP, Cosmos VLMs, image/video-text embeddings, retrieval/ranking, multi-view 3D perception, temporal fusion
ML Engineering	Train/eval pipelines, mixed-precision inference, quantization, pruning, distillation, GPU latency/throughput profiling
Dev Tools	Cursor, Claude Code, OpenAI Codex, Gemini CLI, Git, Docker, JupyterLab, Weights & Biases
Languages	English, Mandarin, Cantonese